

## Lesson Plan:

|                                          |                                                                                                              |                                    |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>Name of the Programme:</b>            | B. Tech. (Hons.)                                                                                             |                                    |
| <b>Title of the Course:</b>              | Probability and Statistics                                                                                   |                                    |
| <b>Course code</b>                       | CS2800                                                                                                       |                                    |
| <b>Semester:</b>                         | 4                                                                                                            |                                    |
| <b>Names of the Course Instructor(s)</b> | Dr. Manjunath S., Dr. T. Hongray, Dr. Bharath M., Dr. Rekha Satish, Dr. Akhilesh K. S., Prof. Amruthesh Bhat |                                    |
| <b>Course credit</b>                     | <b>No. of hours per week (L + T + P)</b>                                                                     | <b>Total no. of Teaching hours</b> |
| 4                                        | 3+1+0                                                                                                        | 45L+15T                            |

| Session | Module No. | Topic                                                  | Pedagogy / activities                  | Pre-reading / reference |
|---------|------------|--------------------------------------------------------|----------------------------------------|-------------------------|
| 1       | 1          | Course overview, applications & motivation             | Lecture / PPT                          |                         |
| 2       | 1          | Sample space, events, relative frequency               | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 1  |
| 3       | 1          | Probability axioms & rules                             | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 1  |
| 4       | 1          | Conditional probability                                | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 2  |
| 5       | 1          | Bayes' Theorem                                         | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 2  |
| 6       | 1          | Independence of events                                 | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 2  |
| 7       | 1          | Simple random walk (1-D)                               | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 3  |
| 8       | 1          | 2-D random walk, Gambler's ruin & fortune distribution | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 3  |
| 9       | 2          | Discrete random variables                              | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 4  |
| 10      | 2          | Expectation & moments of discrete RVs                  | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 4  |
| 11      | 2          | Variance of a sum & covariance                         | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 4  |
| 12      | 2          | Bernoulli trials & Bernoulli distribution              | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 4  |
| 13      | 2          | Binomial distribution — mean & variance                | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 4  |

| Session | Module No. | Topic                                            | Pedagogy / activities                  | Pre-reading / reference  |
|---------|------------|--------------------------------------------------|----------------------------------------|--------------------------|
| 14      | 2          | Binomial applications & approximations           | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 4   |
| 15      | 2          | Geometric distribution & memoryless property     | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 4   |
| 16      | 2          | Negative binomial & Multinomial distributions    | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 4   |
| 17      | 2          | Poisson distribution — properties & examples     | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 4   |
| 18      | 2          | Poisson approximation to Binomial                | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 4   |
| 19      | 2          | Joint distributions & sums of independent RVs    | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 5   |
| 20      | 2          | Correlation coefficient & Chebyshev's inequality | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 5   |
| 21      | 3          | Continuous RVs                                   | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 5   |
| 22      | 3          | Expectation & variance for continuous RVs        | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 4   |
| 23      | 3          | Normal distribution — standardization & Z-scores | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 4   |
| 24      | 3          | Normal distribution applications                 | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 5   |
| 25      | 3          | Exponential distribution & memoryless property   | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 5   |
| 26      | 3          | Gamma distribution — properties & examples       | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 5   |
| 27      | 3          | Relationships among Normal–Exponential–Gamma     | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 5   |
| 28      | 3          | Bivariate distributions — concepts & examples    | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 6   |
| 29      | 3          | Joint density & marginal distributions           | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 6   |
| 30      | 3          | Conditional density & expectations               | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 6   |
| 31      | 3          | Distribution of sums & quotients                 | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 6   |
| 32      | 3          | Bayes' rule in continuous setting                | Lecture + chalk and Talk/ PPT/Activity | Sheldon Ross,Chapter 6   |
| 33      | 4          | Least squares method — curve fitting concepts    | Lecture + chalk and Talk/ PPT/Activity | William Navidi,Chapter 7 |

| Session | Module No. | Topic                                                   | Pedagogy / activities                  | Pre-reading / reference  |
|---------|------------|---------------------------------------------------------|----------------------------------------|--------------------------|
| 34      | 4          | Fitting straight lines — interpretation of coefficients | Lecture + chalk and Talk/ PPT/Activity | William Navidi,Chapter 7 |
| 35      | 4          | Fitting second-degree parabolas                         | Lecture + chalk and Talk/ PPT/Activity | William Navidi,Chapter 8 |
| 36      | 4          | Correlation — Pearson coefficient                       | Lecture + chalk and Talk/ PPT/Activity | William Navidi,Chapter 7 |
| 37      | 4          | Rank correlation — Spearman correlation                 | Lecture + chalk and Talk/ PPT/Activity | William Navidi,Chapter 7 |
| 38      | 5          | Sampling distribution & Central Limit Theorem           | Lecture + chalk and Talk/ PPT/Activity | William Navidi,Chapter 4 |
| 39      | 5          | Null vs alternate hypothesis                            | Lecture + chalk and Talk/ PPT/Activity | William Navidi,Chapter 6 |
| 40      | 5          | z-test                                                  | Lecture + chalk and Talk/ PPT/Activity | William Navidi,Chapter 6 |
| 41      | 5          | t-test                                                  | Lecture + chalk and Talk/ PPT/Activity | William Navidi,Chapter 6 |
| 42      | 5          | Confidence intervals & interpretation                   | Lecture + chalk and Talk/ PPT/Activity | William Navidi,Chapter 6 |
| 43      | 5          | Chi-square test                                         | Lecture + chalk and Talk/ PPT/Activity | William Navidi,Chapter 6 |
| 44      | 5          | Chi-square test — independence of attributes            | Lecture + chalk and Talk/ PPT/Activity | William Navidi,Chapter 6 |
| 45      | 5          | Module recap + problem-solving + summary                | Lecture/ PPT                           |                          |

## Assessment and Evaluation:

| Continuous Internal Evaluation (CIE)                   |                             |
|--------------------------------------------------------|-----------------------------|
| Components of CIE                                      | Weightage of each component |
| CIE 1A(Tutorial): [CO1: 10M]                           | 10                          |
| CIE 1B (Quiz): [CO1: 10M]                              | 10                          |
| CIE 2 (Pen and Paper theory):[CO2: 15M; CO3: 10M; ]    | 25                          |
| CIE 3A (Activity): [CO3: 5M; CO4: 5M; ]                | 10                          |
| CIE 3B (Pen and Paper theory):[CO4: 15M]               | 15                          |
| Semester End Exams (SEE)                               |                             |
| Mode of Exam                                           | Pen and Paper               |
| Weightage of exam: [CO1:5M; CO2:10M; CO3:10M; CO4:5M ] | 30                          |

## Course Outcomes-Programme Outcomes Matrix

| Course Outcomes                                                                                                                                 | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1:Apply basic concepts of probability theory to solve real world problems.                                                                    | 3   | 1   |     |     |     |     |     |     | 2   | 2    |      |      |
| CO2:Apply discrete probability distribution functions to solve real world problems.                                                             | 3   | 2   |     |     |     |     |     |     | 2   | 2    |      |      |
| CO3:Apply continuous probability distribution functions to solve real world problems.                                                           | 3   | 2   |     |     |     |     |     |     | 2   | 2    |      |      |
| CO4:Apply the method of least squares to fit a data set, and interpret the results in the context of the problem at hand.                       | 3   | 2   |     |     |     |     |     |     | 2   | 2    |      |      |
| CO5:Test a given hypothesis using methods like critical value, p-value and chi-square tests for goodness of fit and independence of attributes. | 3   | 3   | 1   | 3   |     |     |     |     | 2   | 2    |      |      |

## Rubrics for evaluation of CIE (separate for each CIE component)

|                          |        |                                                                                                                                                                                                                                 |
|--------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Total Marks</b>       |        | 10                                                                                                                                                                                                                              |
| <b>CIE-1A Components</b> |        | Rubrics for Assessment                                                                                                                                                                                                          |
| CIE-1A                   | Theory | Tutorials will be conducted for 10 marks. Questions will be a mixed of conceptual understanding (L2) questions and problem applications (L3). Evaluation is done according to the scheme of evaluation prepared by the faculty. |

|                          |               |                                                                                                                                                                             |
|--------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Total Marks</b>       |               | 10                                                                                                                                                                          |
| <b>CIE-1B Components</b> |               | Rubrics for Assessment                                                                                                                                                      |
| CIE-1B                   | Activity/Quiz | Poster on application of probability in real scenerios (2 marks for submission on time, 4 marks for presentation and 4 marks for viva)/ Quiz can be conducted for 10 marks. |

|                         |        |                                                                                                                                                                                                                                                   |
|-------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Total Marks</b>      |        | 25                                                                                                                                                                                                                                                |
| <b>CIE-2 Components</b> |        | Rubrics for Assessment                                                                                                                                                                                                                            |
| CIE-2                   | Theory | Pen and paper written test will be conducted for 25 marks. Questions will be a mixed of conceptual understanding (L2) questions and problems applications (L3). Evaluation is done according to the scheme of evaluation prepared by the faculty. |

|                          |          |                                                                                                                                                                                                        |
|--------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Total Marks</b>       |          | 10                                                                                                                                                                                                     |
| <b>CIE-3A Components</b> |          | Rubrics for Assessment                                                                                                                                                                                 |
| CIE-3A                   | Activity | Python/C programming activity. Questions will be given on understanding (L2) and applications (L3) levels (4 marks for code, 2 marks for comments, 2 marks for output, 2 marks for on time submission) |

|                          |        |                                                                                                                                                                                                                                                  |
|--------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Total Marks</b>       |        | 15                                                                                                                                                                                                                                               |
| <b>CIE-3B Components</b> |        | Rubrics for Assessment                                                                                                                                                                                                                           |
| CIE-3B                   | Theory | Pen and paper written test will be conducted for 15 marks. Questions will be a mixed of conceptual understanding (L2) questions and problems applications (L3). Evaluation is done according to the scheme of evaluation prepared by the faculty |

|                       |        |                                                                                                                                                                                                                                                   |
|-----------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Total Marks</b>    |        | 30                                                                                                                                                                                                                                                |
| <b>SEE Components</b> |        | Rubrics for Assessment                                                                                                                                                                                                                            |
| SEE                   | Theory | Pen and paper written test will be conducted for 30 marks. Questions will be a mixed of conceptual understanding (L2) questions and problems applications (L3). Evaluation is done according to the scheme of evaluation prepared by the faculty. |